

Lectures 10b: Hybrid Models

(Slides by Dr. Cook)

Learning Objectives

- Hybrid model
 - How do these models differ from linear and logistic models?
 - Interpretation of coefficients
 - Things to consider

Hybrid Models

Hybrid Models

- **Random effect models**

- Key assumption is that the level-2 error is not correlated with the covariates
- If this assumption is not upheld then the coefficients are biased (e.g. omitted variable bias)

- **Fixed effect models**

- These models do not require the assumption that the level-2 error is not correlated with the covariates
- This model can provide unbiased estimates of the level one variables
- However, fixed effects models focus on level-1 estimates and thus do not provide estimates for the level-2 variables

Hybrid Models

- Combines the benefits of fixed and random effect models
 - Allows us to model level-1 covariates that are not biased by level-2 heterogeneity
 - Allows us to model level-2 time invariant covariates
- Can use this technique with the linear or logit function

Hybrid Models

$$g(\mu_{ij}) = \beta_W(x_{ij} - \bar{x}_i) + \beta_B\bar{x}_i + \gamma c_i + u_i$$

- Within each group, calculate the mean for each independent time-varying variables (X_{ij} 's). The means will represent the between-group differences ($\bar{X}$).
- Then, within each group, subtract the mean for the group from each time-varying variable. These deviations from the group mean will represent the within-group variability.
- Lastly, estimate a random effects model that includes both the means of the variables and the difference-from-the-means variables.
- You can also include time-invariant predictors.

Example 1

- Sample – 1151 teenaged girls interviewed annually for five years, 1979 -1984

[DATA: use “<https://www3.nd.edu/~rwilliam/statafiles/teenpovxt.dta>”, clear]

Outcome Variable

- **POV:** **1** if the girl’s household was in poverty, **0** household not in poverty

Predictor Variables

- **Age:** Age in years at first interview
- **Black:** **1** if Black, **0** if other
- **Mother:** **1** at least 1 child, **0** no children
- **Spouse:** **1** currently live with spouse, **0** if other
- **School:** **1** currently enrolled, **0** not enrolled
- **Hours:** Hours worked week of survey

Example 1: Random Effects Model

xtlogit pov age black mother spouse school hours i.year, re

Random-effects logistic regression
Group variable: id

Number of obs = 5,755
Number of groups = 1,151

Random effects u_i ~ Gaussian

Obs per group:
min = 5
avg = 5.0
max = 5

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -3403.7655

Wald chi2(10) = 266.60
Prob > chi2 = 0.0000

pov	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
age	-.0627952	.0472163	-1.33	0.184	-.1553373	.029747
black	.6093942	.0975653	6.25	0.000	.4181698	.8006186
mother	1.009877	.118372	8.53	0.000	.7778724	1.241882
spouse	-1.171833	.1512544	-7.75	0.000	-1.468286	-.8753802
school	-.1145721	.0990775	-1.16	0.248	-.3087604	.0796163
hours	-.0259014	.0028771	-9.00	0.000	-.0315403	-.0202624
year						
2	.2830958	.1000437	2.83	0.005	.0870138	.4791778
3	.213423	.1040523	2.05	0.040	.0094842	.4173618
4	.2415184	.1090094	2.22	0.027	.0278639	.455173
5	.1447937	.1161395	1.25	0.212	-.0828355	.372423
_cons	-.0045847	.7620829	-0.01	0.995	-1.49824	1.48907
/lnsig2u	.3086358	.1008833			.1109083	.5063634
sigma u	1.166862	.0588584			1.057021	1.288117
rho	.2927197	.0208864			.2535175	.3352612

LR test of rho=0: chibar2(01) = 327.62

Prob >= chibar2 = 0.000

Example 1: Fixed Effect Model

- **Age** and **Black** have dropped out of the model

- Coefficients for **Mother, Spouse, and Hours** are smaller in magnitude with larger standard errors, and the **School** coefficient changed signs

- It is common for coefficients to be different in FE models because it controls for all characteristics of the individual

xtlogit pov age black mother spouse school hours i.year, fe

Conditional fixed-effects logistic regression
Group variable: id

Number of obs = 4,135
Number of groups = 827

Obs per group:
min = 5
avg = 5.0
max = 5

Log likelihood = -1520.1139

LR chi2(8) = 97.28
Prob > chi2 = 0.0000

	pov	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
	age	0	(omitted)				
	black	0	(omitted)				
	mother	.5824322	.1595831	3.65	0.000	.269655	.8952094
	spouse	-.7477585	.1753466	-4.26	0.000	-1.091431	-.4040854
	school	.2718653	.1127331	2.41	0.016	.0509125	.4928181
	hours	-.0196461	.0031504	-6.24	0.000	-.0258208	-.0134714
	year						
	2	.3317803	.1015628	3.27	0.001	.132721	.5308397
	3	.3349777	.1082496	3.09	0.002	.1228124	.547143
	4	.4327654	.1165144	3.71	0.000	.2044013	.6611295
	5	.4025012	.1275277	3.16	0.002	.1525514	.652451

Example 1: Comparing RE and FE Models

```
xtlogit pov age black mother spouse school hours i.year, re
est store random
xtlogit pov age black mother spouse school hours i.year, fe
est store ranfixed
```

```
. hausman fixed random
```

---- Coefficients ----				
	(b)	(B)	(b-B)	sqrt(diag(V_b-V_B))
	fixed	random	Difference	S.E.
mother	.5824322	1.009877	-.427445	.1070274
spouse	-.7477585	-1.171833	.4240749	.0887048
school	.2718653	-.1145721	.3864373	.053781
hours	-.0196461	-.0259014	.0062553	.0012836
year				
2	.3317803	.2830958	.0486845	.0175002
3	.3349777	.213423	.1215547	.029851
4	.4327654	.2415184	.191247	.0411407
5	.4025012	.1447937	.2577075	.0526777

b = consistent under Ho and Ha; obtained from xtlogit
 B = inconsistent under Ha, efficient under Ho; obtained from xtlogit

Test: Ho: difference in coefficients not systematic

```
chi2(8) = (b-B)' [(V_b-V_B)^(-1)] (b-B)
        = 168.18
Prob>chi2 = 0.0000
```

Example 1: Hybrid Model

Recall – We will be combining the fixed and random effects

1. Calculate means of the time-varying predictors for each persons.
2. Calculate deviations from person-specific means.
3. Estimate a RE model with both means and deviations.
4. Test for differences in coefficients.

Example 1: Hybrid Model

```
xthybrid pov age black, use(mother spouse school hours year)
family(binomial) link(logit) clusterid(id) star
Hybrid model. Family: binomial. Link: logit.
```

Variable	model
pov	
R__age	-0.1233*
R__black	0.5707***
W__mother	0.6013***
W__spouse	-0.8234***
W__school	0.2914**
W__hours	-0.0208***
W__year	0.0905**
B__mother	1.0784***
B__spouse	-2.1452***
B__school	-1.3586***
B__hours	-0.0468***
B__year	(omitted)
_cons	2.1895**
var(_cons[id])	
_cons	1.2425***
Statistics	
ll	-3367.4578
chi2	328.2321
p	0.0000
aic	6760.9156
bic	6847.4673

The coefficients are similar to fixed effects model but not the same.

We are interested in the vars that start with R (time invariant) and W (time varying).

If the assumptions of the RE model are true, the coefs for B_ variables (between-group) should equal the coefs for the corresponding W_ vars.

legend: * p<.05; ** p<.01; *** p<.001
Level 1: 5755 units. Level 2: 1151 units.

Example 2: Hybrid Model

```
. webuse nlswork
```

(National Longitudinal Survey. Young Women 14-26 years of age in 1968)

```
. describe idcode hours age msp race
```

variable name	storage type	display format	value label	variable label
idcode	int	%8.0g		NLS ID
hours	int	%8.0g		usual hours worked
age	byte	%8.0g		age in current year
msp	byte	%8.0g		1 if married, spouse present
race	byte	%8.0g	racelbl	race

```
. generate black = race==2 if race!=.
```

```
. generate other = race==3 if race!=.
```

Example 2: Hybrid Model

```
. xthybrid hours age msp black other, clusterid(idcode) se test
```

Hybrid model. Family: gaussian. Link: identity.

Variable		model
hours		
R__black		0.5470
		0.2278
R__other		-0.0404
		0.9396
W__age		-0.0236
		0.0096
W__msp		-1.1661
		0.1529
B__age		0.0552
		0.0200
B__msp		-3.3647
		0.2685
_cons		36.5595
		0.5957
var(_cons[idcode])		
_cons		30.2310
		0.9900
var(e.hours)		
_cons		68.6073
		0.6334
Statistics		
ll		-1.032e+05
chi2		259.5143
p		0.0000
aic		2.065e+05
bic		2.066e+05

legend: b/se

Level 1: 28428 units. Level 2: 4709 units.

Tests of the random effects assumption:

_b[B__age] = _b[W__age]; p-value: 0.0004
_b[B__msp] = _b[W__msp]; p-value: 0.0000

When statistically significant a RE Model is not preferred.

Within- Person

- women in this sample **work fewer hours** in those years in which they are **younger and unmarried** relative to those years in which they are **older and married**, all else being equal.

Between-Person

- A 1-unit increase in age is associated with a .055 increase in work hours, all else being held equal.
- Never married women work about 3 hours less than women who are always married, all else being held equal.

Example 2: Hybrid Model

xthybrid hours age msp black other, clusterid(idcode) se randomslope(age) iterations

```

Mixed-effects GLM                                Number of obs      =      28,428
Family:                                           Gaussian
Link:                                             identity
Group variable:                                idcode

                                                Number of groups   =       4,709

Obs per group:
    min =                                         1
    avg  =                                         6.0
    max  =                                         15

Integration method: mvaghermite                  Integration pts.    =          7

Log likelihood = -102298.83                      Wald chi2(6)       =      201.16
                                                Prob > chi2        =      0.0000
-----
      hours |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-----+-----
      R__black |   .5174951   .2310199     2.24   0.025     .0647044   .9702859
      R__other |  -.1408401   .9487884    -0.15   0.882    -2.000431   1.718751
      W__age   |  -.0142051   .0163478    -0.87   0.385    -.0462462   .0178361
      W__msp   |  -.6070389   .1589173    -3.82   0.000    -.9185111  -.2955667
      B__age   |   .0579946   .0199396     2.91   0.004     .0189137   .0970755
      B__msp   |  -3.363616   .2707035   -12.43   0.000    -3.894185  -2.833047
      _cons    |   36.4563    .5944607    61.33   0.000     35.29118   37.62142
-----+-----
idcode      |
  var(W__age) |   .4968284   .0233867             .4530423   .5448465
  var(_cons)  |   34.09215   1.030189             32.13164   36.17227
-----+-----
  var(e.hours) |   55.18712   .5573331             54.10551   56.29035
-----+-----
LR test vs. linear model: chi2(2) = 5687.80      Prob > chi2 = 0.0000

```

Example 2: Hybrid Model Fit

```
xthybrid hours age msp black other, clusterid(idcode) se
```

(output omitted)

```
estimates store model1
```

```
xthybrid hours age msp black other, clusterid(idcode) se randomslope(age)
```

(output omitted)

```
estimates store model2
```

```
lrtest model1 model2
```

```
Likelihood-ratio test
```

```
(Assumption: model1 nested in model2)
```

```
LR chi2(1) = 1861.97
```

```
Prob > chi2 = 0.0000
```

We are comparing the restrictive model (no random slope) to the unrestrictive model (random slope). The unrestrictive model is preferred.

xthybrid command

- There are some limitations to the `xthybrid` command
 - Factor variable notation (e.g., `i.gender`) is not supported and thus you must create dummy variables yourself.
 - Temporary variables are created but then deleted. As a result, some post-estimation commands (e.g., `predict`) will not work.

Hybrid Model Limitations

- You may not be able to estimate marginal effects correctly.
- Other post-estimation commands available may or may not work correctly.
- Interaction terms may be cumbersome
- Logit and Poisson models may not yield estimates consistent with fixed effect models

Summary

- Hybrid models are incredibly useful but under utilized
- This modelling approach allows you to estimate both within and between effects while obtaining unbiased estimates of the within person estimates
- With linear outcomes the within person estimates will mirror those found in the fixed effects model. However, in logistic models there will be slight deviations
- You can utilize random-intercept or random-slope models
- You can use the *lrtest* to determine the best fitting model